

Denis Leang

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Education

Columbia University <i>M.S. in Data Science</i> Relevant Coursework: Machine Learning, Artificial Intelligence, Computer Vision, Statistical Inference, Exploratory Data Analysis & Visualization, Computer Systems	Sep 2023 - Dec 2024 New York, NY
Telecom Saint-Etienne <i>B.S./M.S. in Computer Science (Diplome d'ingenieur)</i> Relevant Coursework: Data Structures & Algorithms, Database and Information Systems, Operating Systems, Distributed Computing, Cloud Computing, Big Data, Computer Graphics, Probability & Statistics, Optimization & Estimation methods	Sep 2020 - Aug 2023 Saint-Etienne, FRA
Lycee du Parc <i>Preparation for the French national entrance examinations to Grandes Ecoles (CPGE - Mathematics, Physics, Chemistry)</i>	Sep 2017 - Aug 2020 Lyon, FRA

Experience

Columbia University Mailman School of Public Health <i>Graduate Research Assistant</i> - Designed and implemented a RAG AI agent in Python that retrieves and synthesizes public health data from research databases, improving information retrieval efficiency and reducing search time - Containerized the backend API with Docker and deployed it on AWS ECS, improving scalability and reducing deployment times - Fine-tuned OpenAI's Whisper voice recognition model and implemented signal processing techniques, boosting transcription accuracy from 80% to 90%	Mar 2025 - Jan 2026 New York, NY
Analysis Group <i>Data Scientist Intern</i> - Revamped a CRUD patient survey management app built with C# (.NET) and React, enabling more efficient distribution of clinical drug testing surveys by honing workflows and strengthening vendor engagement - Reduced user navigation time by 50% by redesigning the application's structure using scalable React components and optimizing API calls, improving overall user efficiency - Executed Python scripts leveraging the Etherscan API to analyze cryptocurrency transactions across multiple clusters, tracking billions of dollars on Binance and facilitating investigations into suspicious wallet activity	Jun 2024 - Aug 2024 Boston, MA
Bionomous SA <i>Full-Stack Software Engineer Intern</i> - Developed a CRUD client update and support portal written in Java Spring Boot and Vue.js to automate software updates for in-house products, eliminating manual engineer involvement and improving customer experience and release management - Implemented a secure authentication system with Microsoft Azure Active Directory and JWT tokens, boosting login efficiency and strengthening security for users and administrators - Upgraded company products (Java for Android, Flutter/Dart for Android/iOS) by incorporating client-requested functionalities, increasing usability and customer satisfaction	Feb 2023 - Aug 2023 Villaz Saint-Pierre, CHE

Projects

Spotify Dashboard - Built a Spotify listening habits dashboard in React with real-time, dynamically fetched data from the Spotify and Last.fm APIs, visualizing trends through interactive charts - Refined data fetching for real-time music analytics by executing an efficient caching strategy and a CI/CD pipeline with GitHub Actions, reducing memory usage and query time by 4x
Iris Recognition Model - Engineered an iris recognition package with OpenCV and Python, incorporating preprocessing, feature extraction, and LDA-based matching to support one-to-one and one-to-many biometric recognition modes - Achieved a consistent 95% Correct Recognition Rate (CRR) by implementing a Gabor-based multichannel filtering technique, enhancing feature extraction across L1, L2, and Cosine similarity metrics
Electronic Health Record (EHR) Study on Interstitial Lung Disease with Columbia University Irving Medical Center - Conducted data wrangling in Python to harmonize large-scale EHR data across multiple health systems - Fine-tuned Named Entity Recognition (NER) and BERT-based models on medical text to analyze chest CT reports, achieving a 30% improvement in accuracy for identifying drug-disease associations, contributing to enhanced clinical treatment guidelines

Skills

Coding: Proficient in Python, SQL. Experience in Java, C#, TypeScript, Dart.
Technologies: Proficient in Git, Docker, pandas, scikit-learn, Spring Boot, React. Experience in Linux, Azure, AWS, ChromaDB, OpenCV, Tensorflow, PyTorch, Spark, Vue.js, Flutter. Used GCP, Kubernetes, Kafka, MongoDB, Redis, OpenGL, Leaflet, FastAPI.
Languages: French, English, Khmer, German